

In []:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')

# Time Series Decomposition

decomposition = seasonal_decompose(data['Page_Views'], model='additive')

decomposition.plot()
```

```
plt.show()
```

```
# Moving Average
```

```
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
```

```
data[['Page_Views', 'Page_Views_MA']].plot()
```

```
plt.title('Page Views with Moving Average')
```

```
plt.show()
```

```
# Anomaly Detection
```

```
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /  
data['Page_Views'].std()
```

```
anomalies = data[data['Page_Views_Z'].abs() > 3]
```

```
plt.plot(data.index, data['Page_Views'], label='Page Views')
```

```
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
```

```
plt.legend()
```

```
plt.show()
```

